Second-Labeler Packet: DeFi Exploit Categorization

Thank you for helping. You are an independent second labeler for 24 real DeFi exploits. Your labels will be compared with another labeler's to measure agreement. Please work blind:

- Do not look at the project repository, the paper, or anyone else's labels.
- Do not discuss any incident with Sanjay until you have submitted your answer sheet.
- Budget about 5 minutes per incident (about 2 hours total). If you cannot decide, use O.

For each incident, fill one row of the answer sheet (second_labeler_answers.xlsx).

What to decide

Your job is to decide whether each exploit was an oracle-price manipulation (category M) or something else. Two terms matter:

- Price source (S): an on-chain value that reports a price or exchange rate, such as a pool's spot price, a time-weighted average price, a vault's share price, or an exchange rate.
- Consumer (C): a different contract that reads S and uses it to value collateral, mint, redeem, swap, or liquidate.

Category M: oracle-price manipulation. Label M only if ALL four hold:

- M1. An attacker action (possibly funded by a flash loan) moved the value of a price source S away from its fair value, at or before the exploit transaction.
- M2. A different contract C reads S (directly or through a cached value) and uses it to value collateral, mint, redeem, swap, or liquidate.
- M3. The attacker's profit comes from C relying on the moved value.
- M4. S moved because of attacker-controlled flows into S (swaps, deposits, donations, mint or burn, self-trades), or because the attacker exploited an arithmetic flaw in S's own math while a separate consumer still read the result. In that second case, also tick the math_enabled column.

If M does not hold, choose the first that fits, in this order:

- R: a rounding, precision, or invariant-math bug in a pool or curve, with no separate consumer reading a manipulated price.
- A: an accounting, access-control, or logic bug (share accounting, unchecked callbacks, mint bugs) where the price source and its only consumer are the same contract.
- K: an exploit of an MEV bot, a private key, a permission, or governance, unrelated to a consumed price.
- O: the evidence is not enough to decide.

Tie-break: if the price source and its only consumer are the same contract and the flaw is in that contract's own share or rate accounting, label A, even if it looks like manipulation. A donation to a vault whose rate is read by a different lending contract is M.

Evidence rules

- Use at least two independent sources per incident when possible: a protocol or auditor post-mortem plus an independent analysis (for example BlockSec, PeckShield, SlowMist, OpenZeppelin, Halborn, rekt.news).

- The starting links below are suggestions only. You may use any other sources, and you should record every source you actually relied on.
- A proof-of-concept code comment (for example in DeFiHackLabs) alone counts as a single source.
- Optional: view the attack transaction on a block explorer (the Etherscan link) to see which contracts were called.

The 24 incidents (randomized order)

1. Indexed Finance (2021-10-15)

Attack transaction: https://etherscan.io/tx/0x44aad3b853866468161735496a5d9cc961ce5aa872924c5d78673076b1cd95aa
Starting source: https://ndxfi.medium.com/indexed-attack-post-mortem-b006094f0bdc
Starting source: https://blocksecteam.medium.com/the-analysis-of-indexed-finance-security-incident-8a62b9799836

2. WiseLending (2023-10-13)

Attack transaction: https://etherscan.io/tx/0x7ac4a98599596adbf12fffa2bd23e2a2d2ac7e8989b6ea043fcc412a29126555
Starting source: https://twitter.com/bbbb/status/1712841315522638034
Starting source: https://twitter.com/BlockSecTeam/status/1712871304993689709

3. AjnaFinance (2026-08-28)

Attack transaction: https://etherscan.io/tx/0x12dfde527ef62882bfabb64362c9ae0e6bfb628363bd298d0d0956c9a114e4f5
Starting source: https://www.cryptotimes.io/2026/08/29/ajna-v2-loses-775k-on-ethereum-after-attacker-exploits-liquidation-math/
Starting source: https://cryptoticker.io/en/ajna-defi-lending-exploit-immutable-contract/
Starting source: https://www.bitget.com/news/detail/12560605753354

4. MonoX Finance (2021-11-30)

Attack transaction: https://etherscan.io/tx/0x9f14d093a2349de08f02fc0fb018dadb449351d0cdb7d0738ff69cc6fef5f299
Starting source: https://medium.com/monoswap/exploit-post-mortem-33921a779b43
Starting source: https://slowmist.medium.com/detailed-analysis-of-the-31-million-monox-protocol-hack-574d8c44a9c8
Starting source: https://immunebytes.com/blog/monox-hack-incident-nov-30-2021-detailed-analysis/

5. Inverse Finance (2022-06-16)

Attack transaction: https://etherscan.io/tx/0x958236266991bc3fe3b77feaacea120f172c0708ad01c7a715b255f218f9313c
Starting source: https://blocksec.com/blog/our-take-on-the-inverse-finance-security-incident-price-manipulation-attack
Starting source: https://www.halborn.com/blog/post/explained-the-inverse-finance-hack-june-2022
Starting source: https://github.com/yuichiroaoki/inverse-finance-exploit

6. CurveLlamaLend (2026-03-02)

Attack transaction: https://etherscan.io/tx/0xb93506af8f1a39f6a31e2d34f5f6a262c2799fef6e338640f42ab8737ed3d8a4
Starting source: https://www.llamarisk.com/research/llamalend-sdola-long2-post-mortem
Starting source: https://gov.curve.finance/t/llamalend-sdola-long2-post-mortem/11020

7. ZunamiProtocol (2023-08-14)

Attack transaction: https://etherscan.io/tx/0x0788ba222970c7c68a738b0e08fb197e669e61f9b226ceec4cab9b85abe8cceb
Starting source: https://www.certik.com/blog/zunami-protocol-incident-analysis
Starting source: https://www.halborn.com/blog/post/explained-the-zunami-protocol-hack-august-2023
Starting source: https://neptunemutual.com/blog/how-was-the-zunami-protocol-exploited/

8. BalancerBoostedPool2023 (2023-08-27)

Attack transaction: https://etherscan.io/tx/0x2a027c8b915c3737942f512fc5d26fd15752d0332353b3059de771a35a606c2d
Starting source: https://blocksec.com/blog/tiny-rounding-down-big-fund-losses-an-in-depth-analysis-of-the-recent-balancer-incident
Starting source: https://slowmist.medium.com/review-and-recommendations-of-balancer-incident-d2b31b5bd863
Starting source: https://medium.com/immunefi/balancer-rounding-error-bugfix-review-cbf69482ee3d

9. MEVBOTa47b (2022-10-14)

Attack transaction: https://etherscan.io/tx/0x35ecf595864400696853c53edf3e3d60096639b6071cadea6076c9c6ceb921c1
Starting source: https://twitter.com/BlockSecTeam/status/1580779311862190080
Starting source: https://twitter.com/AnciliaInc/status/1580705036400611328

10. USM (2026-08-09)

Attack transaction: https://etherscan.io/tx/0xfae5e751b8ce01457cbb6b529839f24a0cff50faaabcbd0fd02ca0cf559b050e
Starting source: https://www.nominis.io/insights/nominis-monthly-report

11. SecuredFi (2026-09-05)

Attack transaction: https://etherscan.io/tx/0xcd6159860783d8d481cc4bf4dceddc88acdca6bb08f6445525b2904cfd75b1da
Starting source: https://crypto.training/hacks/

12. UwU Lend (2024-06-10)

Attack transaction: https://etherscan.io/tx/0x242a0fb4fde9de0dc2fd42e8db743cbc197ffa2bf6a036ba0bba303df296408b
Starting source: https://slowmist.medium.com/analysis-of-the-uwu-lend-hack-9502b2c06dbe
Starting source: https://www.quillaudits.com/blog/hack-analysis/uwu-lend-hack
Starting source: https://rekt.news/uwulend-rekt

13. ThetanutsIndexVault (2026-06-15)

Attack transaction: https://etherscan.io/tx/0xbba9f138fe39503bfd1aa62932dbd6ab35d37d23d48e4b7bf2988a9d5dc39fec
Starting source: https://www.darknavy.org/web3/exploits/thetanuts-legacy-index-vault-zero-cost-remint/
Starting source: https://www.cryptopolitan.com/hack-deprecated-thetanuts-vault/

14. ResupplyFi (2025-06-26)

Attack transaction: https://etherscan.io/tx/0xffbbd492e0605a8bb6d490c3cd879e87ff60862b0684160d08fd5711e7a872d3
Starting source: https://www.halborn.com/blog/post/explained-the-resupply-hack-june-2025
Starting source: https://ackee.xyz/blog/resupply-hack-analysis/
Starting source: https://rekt.news/resupplyfi-rekt

15. DRLVaultV3 (2025-11-10)

Attack transaction: https://etherscan.io/tx/0xe3eab35b288c086afa9b86a97ab93c7bb61d21b1951a156d2a8f6f5d5715c475
Starting source: https://blog.verichains.io/p/the-drlvaultv3-exploit-a-slippage

16. Cream Finance (2021-10-27)

Attack transaction: https://etherscan.io/tx/0x0fe2542079644e107cbf13690eb9c2c65963ccb79089ff96bfaf8dced2331c92
Starting source: https://immunefi.com/blog/bug-fix-reviews/hack-analysis-cream-finance-oct-2021/
Starting source: https://inspexco.medium.com/cream-finances-incident-analysis-yusd-share-price-manipulation-ebd55c56d8eb
Starting source: https://www.halborn.com/blog/post/explained-the-cream-finance-hack-october-2021

17. Harvest Finance (2020-10-26)

Attack transaction: https://etherscan.io/tx/0x35f8d2f572fceaac9288e5d462117850ef2694786992a8c3f6d02612277b0877
Starting source: https://slowmist.medium.com/slow-mist-analysis-of-harvest-finances-hacked-event-63450b49e6a5
Starting source: https://rekt.news/harvest-finance-rekt
Starting source: https://blockapex.io/harvest-finance-hack-analysis-poc/

18. Makina (2026-01-20)

Attack transaction: https://etherscan.io/tx/0x569733b8016ef9418f0b6bde8c14224d9e759e79301499908ecbcd956a0651f5
Starting source: https://www.certik.com/blog/makina-incident-analysis
Starting source: https://blog.verichains.io/p/analyzing-the-4m-makina-finance-exploit
Starting source: https://makinafi.substack.com/p/post-mortem-january-20th-2026-incident

19. BalancerV1BPool (2026-08-31)

Attack transaction: https://etherscan.io/tx/0x72510b257cc09bde8435b83ac1636f9498ffc353583330600b5b8da43d0d1aff
Starting source: https://news.bitcoin.com/crypto-news/balancer-v1-pool-exploit-234k-rounding-bug/
Starting source: https://blockfence.io/balancer-v1-pool-exploited-for-234k-through-rounding-bug/
Starting source: https://www.cryptotimes.io/2026/08/31/balancer-v1-exploit-hits-unmaintained-code-as-forks-remain-exposed/

20. CompoundUni (2024-02-23)

Attack transaction: https://etherscan.io/tx/0xaee0f8d1235584a3212f233b655f87b89f22f1d4890782447c4ef742b37af58d
Starting source: https://x.com/0xLEVI104/status/1762092203894276481

21. Conic Finance 02 (2023-07-22)

Attack transaction: https://etherscan.io/tx/0x37acd17a80a5f95728459bfea85cb2e1f64b4c75cf4a4c8dcb61964e26860882
Starting source: https://www.certik.com/skynet-report/5FZAakbsUVg1EyHLNw9Qnm-conic-finance-incident-analysis
Starting source: https://www.halborn.com/blog/post/explained-the-conic-finance-hack-july-2023
Starting source: https://medium.com/@ConicFinance/post-mortem-eth-and-crvusd-omnipool-exploits-c9c7fa213a3d

22. EFLeverVault (2022-10-14)

Attack transaction: https://etherscan.io/tx/0x160c5950a01b88953648ba90ec0a29b0c5383e055d35a7835d905c53a3dda01e
Starting source: https://www.web3isgoinggreat.com/?id=earningfarm-exploited-for-971000
Starting source: https://crypto.news/287000-reportedly-stolen-in-hack-targeting-aaves-earning-farm/

23. BalancerV2 (2025-11-03)

Attack transaction: https://etherscan.io/tx/0x6ed07db1a9fe5c0794d44cd36081d6a6df103fab868cdd75d581e3bd23bc9742
Starting source: https://www.openzeppelin.com/news/understanding-the-balancer-v2-exploit
Starting source: https://research.checkpoint.com/2025/how-an-attacker-drained-128m-from-balancer-through-rounding-error-exploitation/
Starting source: https://blocksec.com/blog/in-depth-analysis-the-balancer-v2-exploit

24. TOPBPool (2026-06-09)

Attack transaction: https://etherscan.io/tx/0x967aa34c69b7775c718545c7f94d92e965eb5fc553c0f27f6f1a9c65c93ac156
Starting source: https://crypto.news/token-of-power-exploit-drains-1-58m-from-balancer-pool/
Starting source: https://beincrypto.com/millions-vanish-tiny-tokens-governance-top/
Starting source: https://www.cryptotimes.io/2026/06/10/one-vote-1-58m-gone-top-token-hit-by-alleged-governance-attack/